

FINAL PROJECT

Hotel Booking Cancellation Specialist

Binary Classification - Hotel Reservation Cancellation

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✓ 1. Data Understanding

```
# 1.1 Import Library

# Analisis Data
import pandas as pd
import numpy as np

# Visualisasi
import matplotlib.pyplot as plt
import seaborn as sns

# Machine Learning
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor, plot_tree
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

# Config Visual
sns.set_theme(style="whitegrid")

# Fungsinya :
# - Mengimpor library analisis data (pandas, numpy)
# - Membuat visualisasi (matplotlib, seaborn)
# - Menyiapkan library machine learning dari scikit-learn
# - Mengatur tampilan visual agar rapi dan mudah dibaca
```

```
# 1.2 Dataset Preparation

# Load dataset
url = "https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-01-01/tidytuesday.csv"
df = pd.read_csv(url)
print("\n=====")

# Ukuran dataset
print("Ukuran Dataset :", df.shape)
print("\n=====")

# Data pertama beberapa
display(df.head())
print("\n=====")

# Overview dataset
df.info()

# Dataset diambil langsung online menggunakan URL
# Kemudian pengecekan awal menggunakan:
# - shape      → melihat ukuran dataset
# - head()     → melihat beberapa data pertama
# - info()     → melihat jumlah kolom dan baris,
#               mengetahui tipe data setiap fitur,
#               identifikasi missing value,
#               membantu menentukan strategi preprocessing
# Load dan pantau overview seperti berkenalan dengan dataset
```



```
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```

```
Ukuran Dataset : (119390, 32)
```

1.3 Missing Value

```
missing_values = df.isnull().sum()
```

```
missing_table = pd.DataFrame({
    'Missing Value': missing_values,
    'Persentase (%)': round((missing_values / len(df)) * 100, 2)
})
```

```
missing_table = missing_table[missing_table['Missing Value'] > 0]
```

```
display(missing_table.sort_values(by='Missing Value', ascending=False))
```

```
# Penting diperiksa karena:
```

```
# - Dapat menurunkan performa model
```

```
# - Menyebabkan bias analisis
```

```
# - Harus dihandle dalam tahap data cleaning
```

```
# Pantau output missing value (tersaji dalam bentuk tabel jumlah dan persentase)
```

```
=====
```

	Missing Value	Persentase (%)	
company	112593	94.31	
agent	16340	13.69	
country	488	0.41	
children	4	0.00	
arrival_date_year	119390	100.00	

1.4 Duplicated Data

```
duplicate_count = df.duplicated().sum()
print(duplicate_count)
```

```
# Kondisi dimana dataset memiliki satu atau beberapa value yang sama persis
```

```
# Penting diperiksa karena dapat menyebabkan:
```

```
# - Bias distribusi data
```

```
# - Model terlalu menghafal pola tertentu
```

```
# - Ketidakseimbangan informasi
```

```
# Pantau jumlah data duplikat sebelum preprocessing.
```

```
31794 previous_cancellations 119390 non-null int64
18 previous_bookings_not_canceled 119390 non-null int64
```

1.5 Separate Categorical & Numerical Columns

```
categorical_cols = df.select_dtypes(include='object').columns
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
print("=== Categorical Columns ===")
print(categorical_cols)
print("\n=== Numerical Columns ===")
print(numerical_cols)
```

```
# Penting untuk:
# - Encoding data kategorikal
# - Scaling data numerikal
# - Analisis distribusi
# - Deteksi outlier

# Secara umum bedakan dari Dtype pada step 1.1:
# - object → column kategorikal
# - int64 / float64 → column numerikal
```

```
=== Categorical Columns ===
Index(['hotel', 'arrival_date_month', 'meal', 'country', 'market_segment',
      'distribution_channel', 'reserved_room_type', 'assigned_room_type',
      'deposit_type', 'customer_type', 'reservation_status',
      'reservation_status_date'],
      dtype='object')

=== Numerical Columns ===
Index(['is_canceled', 'lead_time', 'arrival_date_year',
      'arrival_date_week_number', 'arrival_date_day_of_month',
      'stays_in_weekend_nights', 'stays_in_week_nights', 'adults', 'children',
      'babies', 'is_repeated_guest', 'previous_cancellations',
      'previous_bookings_not_canceled', 'booking_changes', 'agent', 'company',
      'days_in_waiting_list', 'adr', 'required_car_parking_spaces',
      'total_of_special_requests'],
      dtype='object')
```

1.5.1 Categorical Columns

```
for col in categorical_cols:
    print(f"{col}: {df[col].nunique()}")

# Jumlah kategori dalam setiap column
# Fungsinya :
# - Melihat cardinality (jumlah nilai unik) fitur kategorikal
# - Menentukan teknik encoding yg pastinya berbeda-beda
# - Mengidentifikasi fitur dengan kategori terlalu banyak
# Pantau ini sangat penting sebelum encoding
```

```
hotel: 2
arrival_date_month: 12
meal: 5
country: 177
market_segment: 8
distribution_channel: 5
reserved_room_type: 10
assigned_room_type: 12
deposit_type: 3
customer_type: 4
reservation_status: 3
reservation_status_date: 926
```

1.5.2 Numerical Columns

```
display(df.describe().T)

# Statistik deskriptif (mendeteksi numerical columns dan menyajikan ringkasan statistik)
```

```
# Fungsinya :
# - Melihat distribusi data
# - Mengetahui nilai minimum dan maksimum
# - Melihat indikasi outlier
# - Mengetahui rata-rata dan standar deviasi
```

	count	mean	std	min	25%	50%
is_canceled	119390.0	0.370416	0.482918	0.00	0.00	0.00
lead_time	119390.0	104.011416	106.863097	0.00	18.00	69.00
arrival_date_year	119390.0	2016.156554	0.707476	2015.00	2016.00	2016.00
arrival_date_week_number	119390.0	27.165173	13.605138	1.00	16.00	28.00
arrival_date_day_of_month	119390.0	15.798241	8.780829	1.00	8.00	16.00
stays_in_weekend_nights	119390.0	0.927599	0.998613	0.00	0.00	1.00
stays_in_week_nights	119390.0	2.500302	1.908286	0.00	1.00	2.00
adults	119390.0	1.856403	0.579261	0.00	2.00	2.00
children	119386.0	0.103890	0.398561	0.00	0.00	0.00
babies	119390.0	0.007949	0.097436	0.00	0.00	0.00
is_repeated_guest	119390.0	0.031912	0.175767	0.00	0.00	0.00
previous_cancellations	119390.0	0.087118	0.844336	0.00	0.00	0.00
previous_bookings_not_canceled	119390.0	0.137097	1.497437	0.00	0.00	0.00
booking_changes	119390.0	0.221124	0.652306	0.00	0.00	0.00
agent	103050.0	86.693382	110.774548	1.00	9.00	14.00
company	6797.0	189.266735	131.655015	6.00	62.00	179.00
days_in_waiting_list	119390.0	2.321149	17.594721	0.00	0.00	0.00
adr	119390.0	101.831122	50.535790	-6.38	69.29	94.575
required_car_parking_spaces	119390.0	0.062518	0.245291	0.00	0.00	0.00
total_of_special_requests	119390.0	0.571363	0.792798	0.00	0.00	0.00

2. Data Cleaning & Preparation

```
# 2.1 Create Working Dataset
```